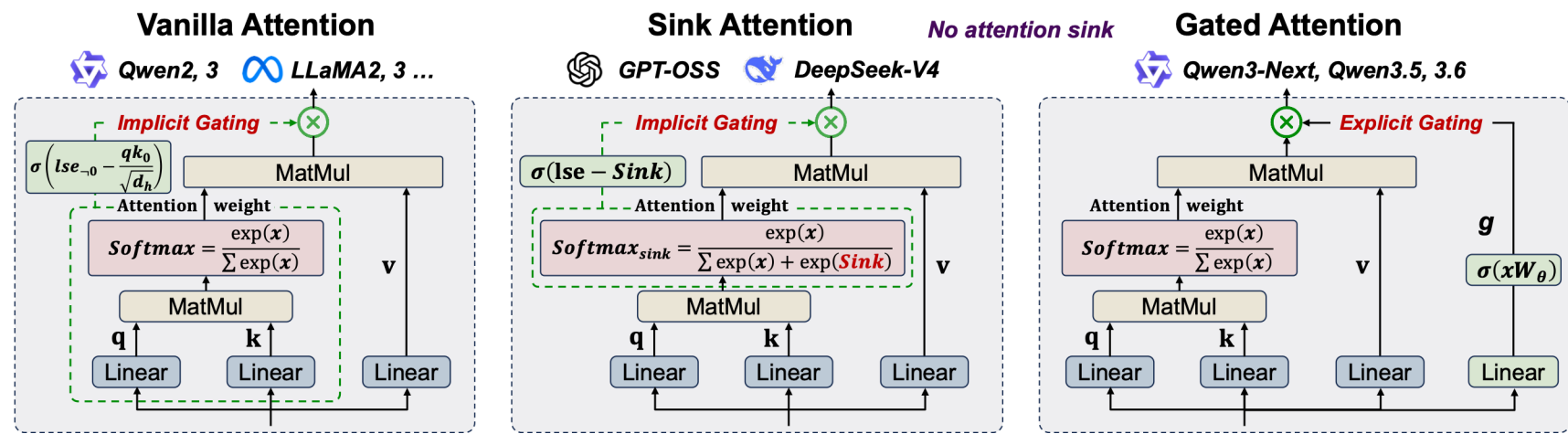


Attention Sink Forges Native MoE in Attention Layers: Sink-Aware Training to Address Head Collapse

Zizhuo Fu^{1,2} Wenxuan Zeng¹ Runsheng Wang² Meng Li^{1,2}

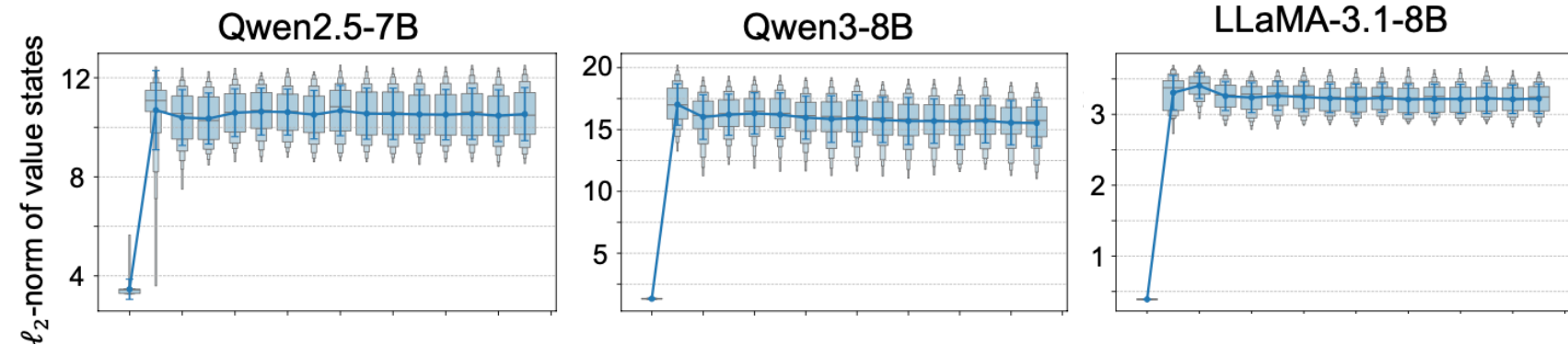
¹Institute for Artificial Intelligence, Peking University ²School of Integrated Circuits, Peking University

Attention Sink Forges Native MoE in Attention Layers

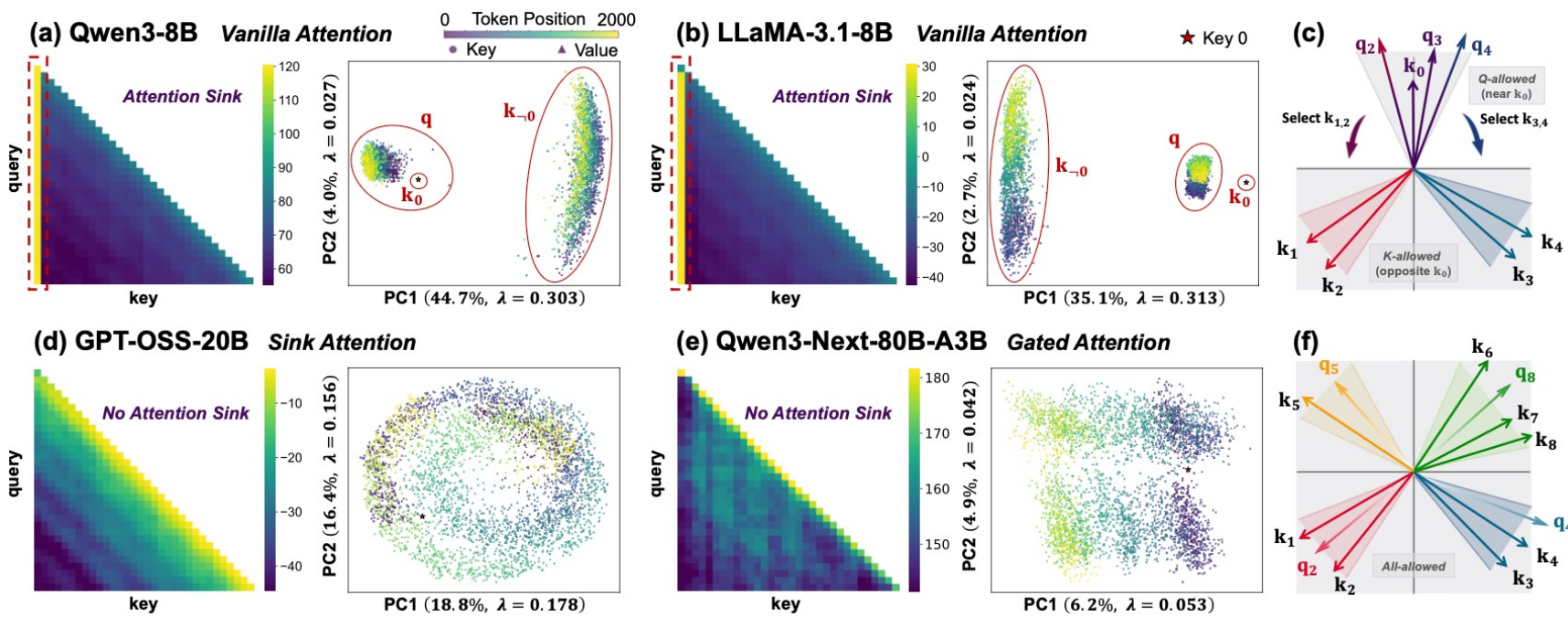


□ **Vanilla Attention** used in most open-source models. **Sink Attention** used in GPT-OSS and DeepSeek-V4 with a learnable bias sink added to the softmax denominator. **Gated Attention** used in Qwen3-Next with a head-wise gating factor computed via sigmoid activation.

Value Drain - l_2 -norm of token values



Visualization of attention patterns and query-key geometry



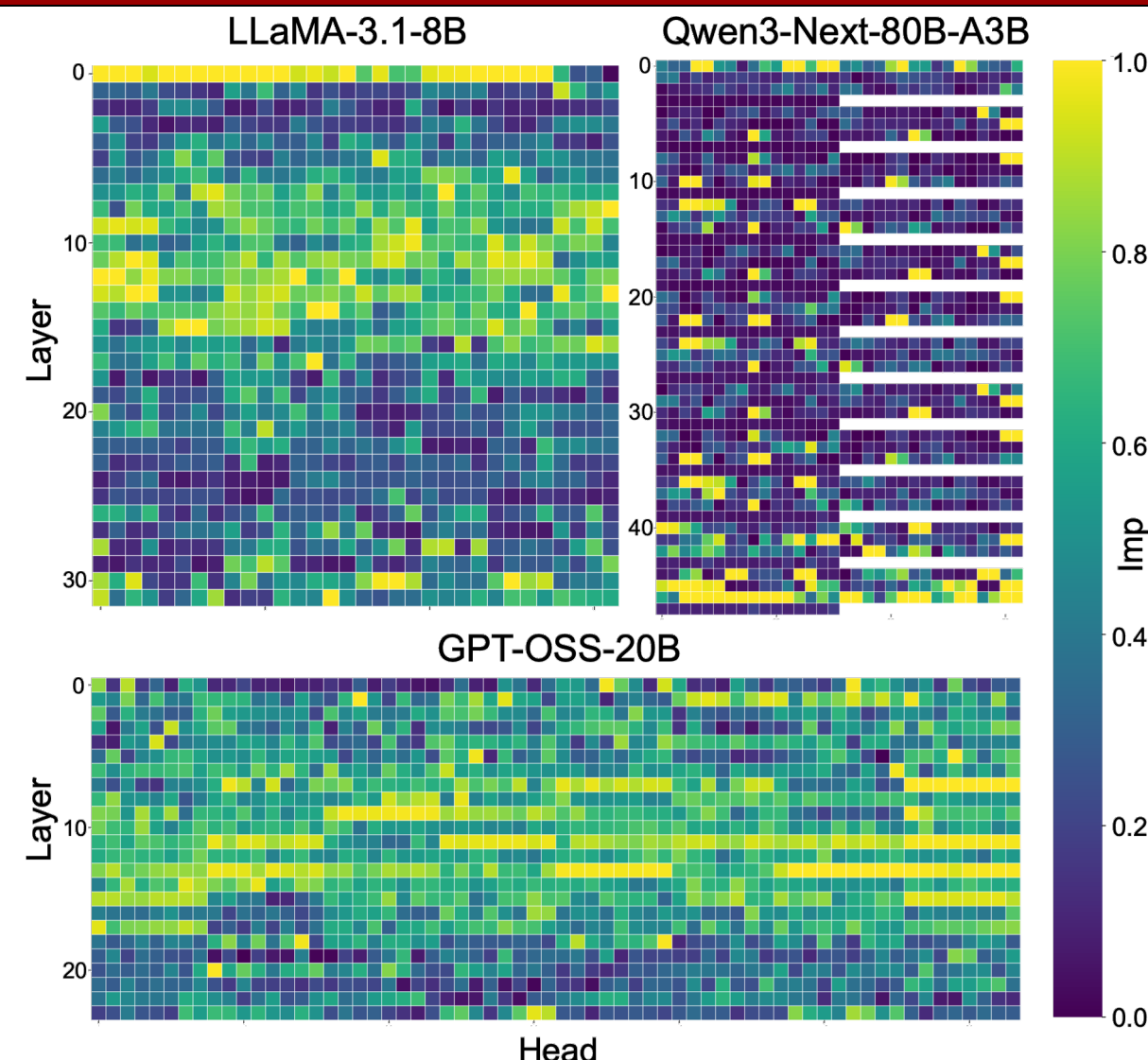
Implicit Gating in Vanilla and Sink Attention

$$O_t^{l,h} = \sum_{i \neq \text{sink}} A_{t,j}^{l,h} \cdot v_j^{l,h} = (1 - A_{t,\text{sink}}^{l,h}) \cdot \sum_{j \neq \text{sink}} \tilde{A}_{t,j}^{l,h} \cdot v_j^{l,h}$$

$$G_t^{l,h} = 1 - A_{t,\text{sink}}^{l,h}$$

□ $\tilde{A}_{t,j}^{l,h}$ denotes the re-normalized softmax attention weights over non-sink tokens

Head collapse - Head importance scores



$$\text{Imp}^{l,h} = \frac{1}{|\mathcal{T}|} \sum_{(i,t) \in \mathcal{T}} G_{i,t}^{l,h}$$

□ Certain heads consistently exhibit high importance scores, while many others show very low activation levels

Sink-Aware Auxiliary Load Balancing Loss

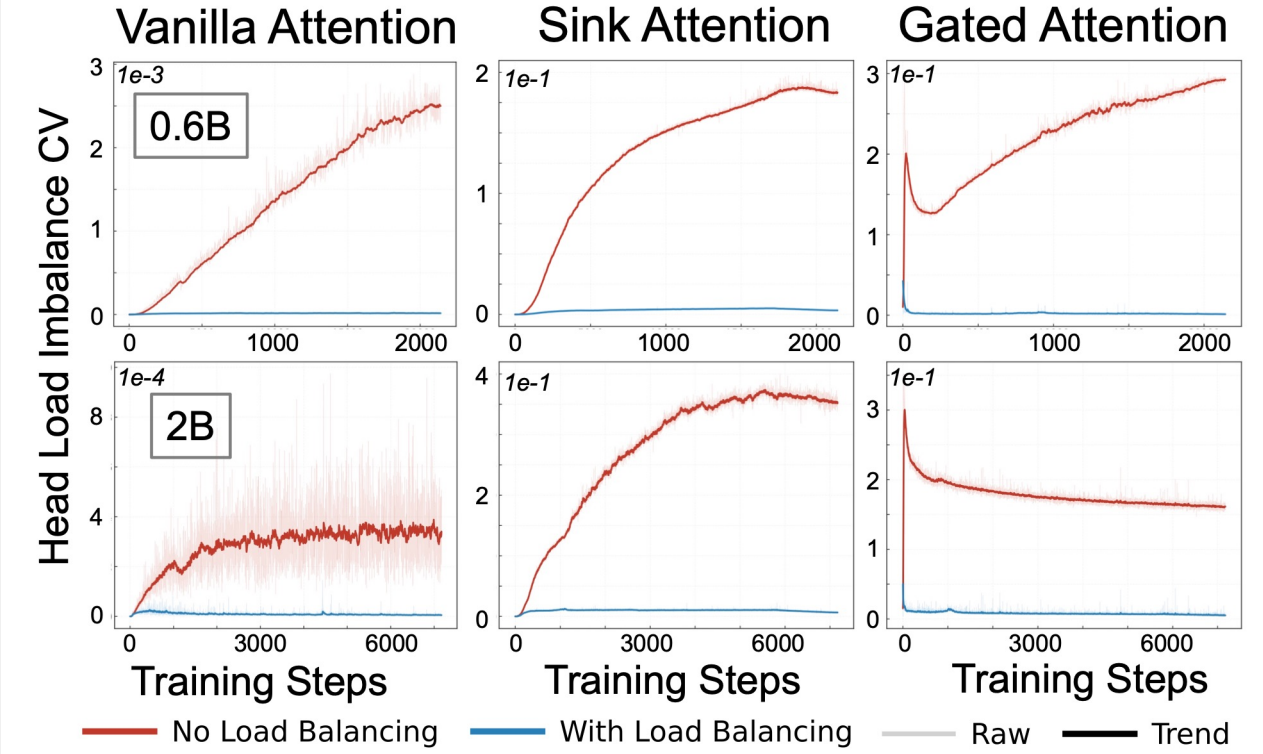
□ **Auxiliary Loss for Training from Scratch**

$$\mathcal{L}_{\text{aux}} = \lambda \sum_{l=1}^{N_L} N_h \left[\text{CV}(\{\text{Imp}^{l,h}\}_{h=1}^{N_h}) \right]^2 \quad \mathcal{L} = \mathcal{L}_{\text{base}} + \mathcal{L}_{\text{aux}}$$

□ **Adaptation to Fine-Tuning Existing Models**

$$\mathcal{L}_{\text{aux}} = \lambda \sum_{l=1}^{N_L} (N_h - m) \left[\text{CV}(\{\text{Imp}^{l,h} : h \notin \mathcal{T}_m^l\}) \right]^2 \quad \mathcal{T}_m^l = \arg \text{top}_m \{\text{Imp}^{l,h}\}_{h=1}^{N_h}$$

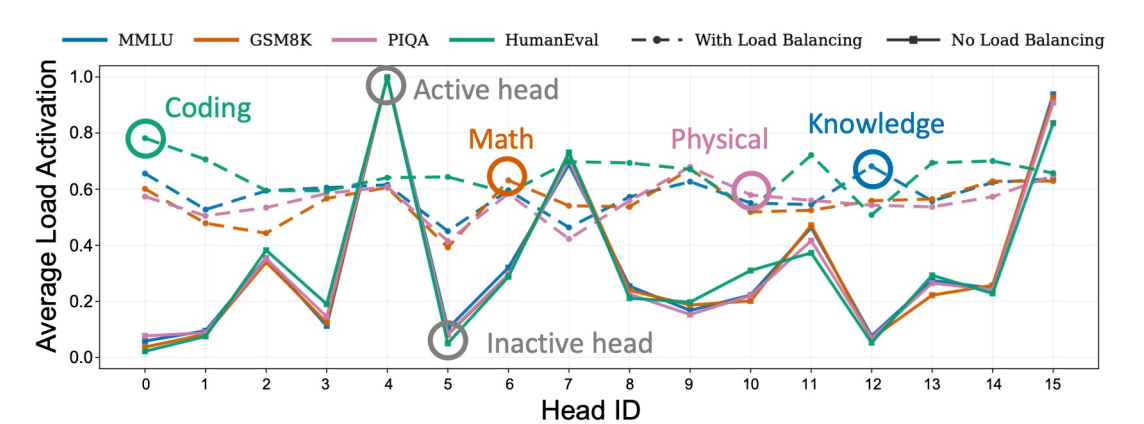
Head load imbalance during training



Experiments & Results

Attention Mechanism	Training Loss	Validation BPP (↓)	MMLU	GSM8K	HumanEval	ARC-E	ARC-C	HellaSwag	PIQA	OpenBookQA	BoolQ	Winogrande	Average
Model Size = 0.6B													
Vanilla	base	0.8152	28.81	5.31	6.10	63.80	32.85	44.70	68.39	34.00	57.15	54.30	39.54
	base+aux	0.8123	32.60	5.84	9.76	65.28	34.47	45.04	68.44	34.40	58.20	54.20	40.82 _(.28)
Sink	base	0.8123	33.09	5.38	7.93	63.30	33.78	44.44	68.11	34.20	60.45	52.40	40.31
	base+aux	0.8116	33.23	5.84	9.76	63.80	34.13	44.81	69.53	36.40	60.76	52.80	41.11 _(.40)
Gated	base	0.8176	31.31	5.53	6.71	63.97	34.89	43.42	68.06	35.80	59.40	55.17	40.43
	base+aux	0.8121	33.24	5.31	9.15	64.86	35.24	44.76	68.72	37.00	60.40	54.80	41.35 _(.02)
Model Size = 1B													
Vanilla	base	0.7591	36.08	6.22	10.98	68.56	38.82	52.13	72.09	37.40	47.06	56.59	42.59
	base+aux	0.7562	37.65	7.43	11.59	68.80	38.99	52.85	72.40	38.00	48.10	57.69	43.55 _(.76)
Sink	base	0.7583	36.58	6.67	5.49	68.73	40.35	52.63	73.40	36.40	58.13	54.30	43.27
	base+aux	0.7567	35.76	7.13	7.32	68.81	40.20	52.80	73.80	36.80	58.60	56.27	43.75 _(.48)
Gated	base	0.7572	36.48	7.73	6.71	67.25	38.73	52.40	70.78	36.00	54.12	56.67	42.69
	base+aux	0.7547	36.73	8.64	9.76	68.35	41.98	53.29	71.54	38.00	54.80	56.75	43.98 _(.29)
Model Size = 2B													
Vanilla	base	0.7243	39.40	9.33	6.71	72.26	41.97	58.26	73.72	37.60	58.96	56.43	45.46
	base+aux	0.7198	41.48	9.63	7.93	72.80	43.09	59.66	74.10	39.40	60.76	59.43	46.83 _(.37)
Sink	base	0.7226	40.46	10.24	3.66	73.06	44.28	58.62	75.24	39.60	59.80	58.56	46.35
	base+aux	0.7202	40.90	10.61	6.10	73.40	44.80	59.22	74.65	39.40	57.40	58.48	46.50 _(.15)
Gated	base	0.7232	38.48	9.63	10.98	72.60	44.36	59.09	74.48	37.40	60.21	55.80	46.30
	base+aux	0.7197	38.98	10.69	11.59	73.20	44.40	59.52	77.20	39.80	61.04	60.60	47.70 _(.40)

□ **Evaluation results of 0.6B, 1B, and 2B models trained from scratch with different attention mechanisms and loss functions**



□ **Head load activation across different datasets trained with and without the auxiliary load balancing loss**